KCPC Week 5 Reading
Graphs II: Dijkstra and 0–1 BFS

Follow this guide alongside the slides and problem sheet. DFS and BFS are covered in *Week 4 Reading*, which this workshop builds on, so this guide only adds what is new: priority queues, shortest paths with non-negative weights (Dijkstra), the relaxation invariant, and 0–1 BFS. For the original research behind these algorithms, see *Week 5 Papers*.

I. Core algorithmic reading

Guide to Competitive Programming (Antti Laaksonen-style community PDF)

https://duoblogger.github.io/assets/pdf/memonvyftw/guide-t-cp.pdf

- **Dijkstra's Algorithm, p. 89:** relaxation with a priority queue for non-negative weights; the basis for Problems 1–2, 4–5 and 9–15.
- **Bellman–Ford, p. 88 (further reading):** the negative-weight alternative the slides warn about; not required for this sheet.
- **Floyd–Warshall, p. 92 (further reading):** all-pairs shortest paths; optional context for why we only need single-source methods here.

II. 0–1 BFS

CP-Algorithms

https://cp-algorithms.com/graph/01_bfs.html
A concise explanation of 0–1 BFS, including its correctness argument and time complexity analysis. Read it before attempting the deque-based problems: 3 (arrow grid), 6 (obstacle removal), 8 (Wizard in Maze) and 15 (Stronger Takahashi). Compare its dial/bucket viewpoint with the slides' "two adjacent distance levels" invariant, and with Dial (1969) in *Week 5 Papers*.

III. Formal textbook reference

Introduction to Algorithms, 3rd Edition (Cormen, Leiserson, Rivest, Stein)

https://www.cs.mcgill.ca/~akroit/math/compsci/Cormen%20Introduction%20to%20Algorithms.pdf
Page numbers below refer to the 3rd edition, as in *Week 4 Reading*.

- **Priority queues, p. 162 (§6.5):** the heap operations behind the slides' lazy min-heap implementation.
- **Dijkstra's Algorithm, p. 658 (§24.3):** the formal proof of the settling invariant the slides rely on, including why negative weights break it.
- **Bellman–Ford, p. 651 (§24.1, further reading):** the correct tool when negative weights appear.
- **Floyd–Warshall, p. 693 (§25.2, further reading):** optional all-pairs perspective.

Suggested order

Start with the DFS and BFS reading in *Week 4 Reading*. Then read the Dijkstra section of the competitive programming guide (p. 89) alongside the slides' worked traces, and solve Problems 1–3. Read the CP-Algorithms 0–1 BFS page, then attempt Problems 3, 6, 8 and 15. Use CLRS §6.5 and §24.3 for the heap operations and the formal settling invariant when writing up your own reasoning, and consult Bellman–Ford and Floyd–Warshall only as further reading. For original sources and historical context, see *Week 5 Papers*.